

APPROACH TO THE PATIENT WITH ABNORMAL SKIN PHOTOSENSITIVITY

Clinical Features

Patients with abnormal skin photosensitivity typically present in one of three ways: sporadic eruptions, persistent eruptions, or, less commonly, erythroderma—almost always affecting habitually light-exposed areas of the skin. In sporadic cases, patients often self-attribute the eruption to sun exposure. However, a detailed history is essential to confirm the role of light and to establish an accurate diagnosis.

Key patient-related factors include: - Age at onset - Gender - Family history of photosensitivity - Personal history of sunlight sensitivity - Occupation and outdoor leisure activities - Use of systemic or topical drugs and chemicals

Important characteristics of the eruption to assess: - Seasonal pattern (e.g., spring/summer onset) - Time between light exposure and eruption onset - Duration of the reaction - Morphology and distribution of lesions - Whether lesions develop after exposure through window glass or clothing (suggesting UVA involvement) - Presence of systemic symptoms - Patient's own description of lesion appearance

Diagnostic Clues Based on Clinical Patterns

- **Polymorphic light eruption (PMLE):** Most common in young women; lesions typically appear hours after exposure and resolve over days. Morphology varies but often includes small or large, red or skin-colored, itchy papules or plaques, sometimes confluent, affecting only certain sun-exposed sites.

- **Actinic prurigo (AP):** Occurs in women and girls; often has a familial component.
- **Hydroa vacciniforme (HV):** Affects children; characterized by vesicles or bullae that heal with pox-like scarring.
- **Xeroderma pigmentosum (XP):** Presents in childhood; extreme photosensitivity with increased risk of skin cancer. Lesions do not develop through glass (UVB-dependent).
- **Erythropoietic protoporphyria (EPP):** Typically begins in childhood; patients report intense burning pain upon light exposure, often without immediate visible changes. Prolonged exposure may cause non-blanching, firm, pink or colorless swelling, occasionally with blisters.
- **Chronic actinic dermatitis (CAD):** More common in elderly males or younger individuals with atopic eczema; often affects those with significant outdoor exposure. May worsen despite sunscreen use—raising suspicion for photoallergy, though exacerbation can occur without allergy in PMLE.

Temporal Patterns of Reaction

- **Immediate reactions (minutes):** Resolution within 1–2 hours suggests solar urticaria (SU), or drug-induced photosensitivity (e.g., amiodarone).
- **Delayed onset (20 minutes to several hours):** Resolution over days points to PMLE, HV, EPP, XP, subacute cutaneous lupus, other photo-exacerbated dermatoses, or drug photosensitivity (e.g., thiazides).

Systemic Symptoms

Systemic malaise is uncommon in PMLE, HV, SU, and cutaneous lupus.

Light Transmission Clues

Development of lesions after exposure through window glass or clothing indicates a UVA (or visible light) component, as UVA penetrates glass. This feature is present in most photodermatoses except XP, which is primarily UVB-mediated.

Photo-Exacerbated Dermatoses

In photo-exacerbated conditions, the eruption typically begins or worsens first at sites commonly affected by the underlying disorder (e.g., face in seborrheic eczema), and may later spread to other or all exposed areas. In photo-exacerbated seborrheic eczema, patients may experience only an unpleasant sensation (e.g., burning, stinging) at exposed sites, without obvious rash.

Management

Treatment focuses on: - Minimizing ultraviolet radiation (UVR) exposure - Wearing protective clothing - Applying high-protection-factor, broad-spectrum sunscreens - Aggressive management of the underlying dermatosis—even if mild or subclinical

Specific measures: - Daily use of anti-eczema shampoo in seborrheic eczema often prevents photosensitivity entirely. - If conservative measures fail, low-dose phototherapy (as used for PMLE) may be beneficial in photo-exacerbated atopic eczema, seborrheic eczema, or psoriasis. - **Contraindication:** Phototherapy should be avoided in cutaneous lupus and dermatomyositis due to the risk of systemic disease aggravation.

Special Cases

- **Photo-exacerbated acne:** Often requires systemic treatment with oral isotretinoin.